

A FAIL-CLOSED CERTIFICATION FRAMEWORK FOR CYCLIC N-ARY LAWS, PROJECTORS, AND MULTIREOLUTION TENSOR STRUCTURE

ELIUTH CHAVERO JASSO

ABSTRACT. We replace a 14-block legacy audit script’s averaging-based scoring system (WARN scores 0.5, N/A scores 1.0, no FAIL state exists) with a fail-closed typed-gate taxonomy: ten states from `STRUCTURAL_IDENTITY_PASS` through `EXACT_CERTIFICATE`, combined by minimum across eight critical gates, never by average. Applying it to the legacy evidence and to a from-scratch v18 reimplementation, we find: every one of 19 unique historical runs is capped at `EMPIRICAL_SCREENING_PASS` (all `eval_mode=screening`, including runs named `*_CERT_*`); the “coherent dynamic curvature” commutator explanation (block B) performs *worse than the trivial zero predictor* in all 15 real checkpoints with block-B data, and a controlled ablation matrix identifies the mechanism as genuine conflict with an associator/GJI-family objective, not insufficient capacity, gradient starvation, or scale imbalance alone; an independently-trained three-resolution interscale transport experiment (blocks E, J, M) finds no persistence signal beyond a single unexplained per-mode anomaly; and a methodological error caught by our own required negative control — comparing subspace bases via free-unitary Procrustes, which is vacuous because the unitary group acts transitively on same-size orthonormal frames — is documented and fixed with principal angles. A follow-up 416-cell sweep across blocks A/G/H/N (Section 4) extends this to a real grid rather than single configurations, and answers a concrete scheduling question: GPU never overtakes CPU for these blocks anywhere in $n \in \{12, 24, 48, 96\}$, a stable $\sim 3\times$ CPU advantage throughout, not a small-scale artifact. No block in this pass reaches a state stronger than `EMPIRICAL_SCREENING_PASS` or `STATISTICALLY_VALIDATED_PASS` for any scientific (as opposed to purely algebraic/structural) claim, and no process here self-issues a final certification: the ceiling is `PASS_A_TO_N_PARTIAL_CERTIFICATION`, pending human review.

1. WHY THE LEGACY SCORING SYSTEM CANNOT BE TRUSTED

The legacy script (`seion_master_audit_A_to_N_v17_blackwell_repro_fix.py`, 2653 lines, hashed and archived at `spectral/legacy/v17/`) assigns each of 14 blocks one of $\{\text{PASS}, \text{PASS_STRONG}, \text{WARN}, \text{WARN_GEOMETRIC}, \text{N/A}\}$ and computes `master_score` as $100\times$ the mean of per-block scores, where $\text{WARN} \mapsto 0.5$ and $\text{N/A} \mapsto 1.0$. There is no FAIL state anywhere in the vocabulary. Consequently a block that never even ran (no high-resolution scale configured) scores full credit, and a block that failed outright scores half credit, not zero.

We ingested all 19 unique historical runs (9 logged in `REPRO_SUMMARIES_A_TO_M_VALIDATED.txt`, 18 on-disk run directories, 8 overlapping) and reconstructed their resume lineage. All 9 logged runs share `seed=3` and are one continuous non-strict-resume chain (`restore_rng=false`, `strict_resume=false` throughout) rooted at a single bootstrap — not nine independent trials. Every run, without exception, has `eval_mode="screening"`, including every directory named `*_CERT_*`. Two distinct `script_sha256` values appear across the lineage: the script currently in the repository matches 8 of 9 logged runs but not the earliest bootstrap run, meaning the script changed mid-campaign without this being tracked anywhere in the legacy artifacts.

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Spectral A-N legacy-audit track (`SPECTRAL_LEGACY_TRACK`, additive scope class, `claims/scope_registry_v4.yaml`). Evidentially separate from the canonical finite-algebra core (`CANONICAL_FINITE_CORE`); no claim below certifies or is certified by that track without a proved connecting theorem. Track T (projected n-ary tree mathematics) is explicitly out of scope, deferred by an accepted mid-campaign checkpoint.

2. THE TYPED-GATE TAXONOMY

We define ten typed states (full definitions and non-implications in `spectral/certification_v18/GATE_TAXONOMY.md`): `STRUCTURAL_IDENTITY_PASS`, `NUMERICAL_SANITY_PASS`, `EMPIRICAL_SCREENING_PASS`, `STATISTICALLY_VALIDATED_PASS`, `VALIDATED_NUMERICAL_CERTIFICATE`, `EXACT_CERTIFICATE`, `WARN`, `FAIL`, `NOT_APPLICABLE`, and `NOT_CERTIFIABLE_AS_DEFINED`. These are not a numeric scale. Eight critical gates (`projector_gate`, `algebra_gate`, `dynamic_explanation_gate`, `interscale_gate`, `gauge_gate`, `persistence_gate`, `reproducibility_gate`, `mathematical_proof_gate`) each take the *minimum* (weakest, never averaged) status over their contributing blocks. A code-level enforcement point (`gates.py:assign_block_status`) raises `ScreeningCertificateViolation` if a screening-mode run is ever assigned a certificate-tier status — a hard error, not a silent downgrade. No code path in this suite can emit `PASS_A_TO_N_FULL_CERTIFICATION`; the ceiling is `PASS_A_TO_N_PARTIAL_CERTIFICATION`, since full certification requires human review this process cannot self-issue.

3. BLOCK-BY-BLOCK RESULTS

3.1. **A — projector validity.** $P = UU^*$ is idempotent and self-adjoint to machine precision for any orthonormal U (5 seeds, plus an exact $n = 2$ case at tolerance 10^{-14}). This is a `STRUCTURAL_IDENTITY_PASS` by construction and is never reported as evidence about the learned subspace’s relevance.

3.2. **B — dynamic commutator explanation.** The candidate explanation $C_\theta = U\Phi U^*\Delta K - K\Delta U\Phi^*U^*$ for $[\Delta, P] = K\Delta P - P\Delta K$ (exact identity, both sides rank $\leq 2r$) is not capacity-limited in isolation: training U against *only* the commutator-unexplained loss drives the residual to $\sim 10^{-9}$ across 5 seeds, matching a closed-form optimal- Φ ceiling exactly. Yet in every one of 15 real historical checkpoints, `coherence_ratio` ≤ 0 : C_θ performs at or worse than predicting zero. A seven-regime ablation matrix isolates the mechanism (Table 1): freezing the law and training only the projector reaches unexplained-rel 0.99 (barely moves); freezing the projector and training only the law reaches unexplained-rel exactly 0 even with U left at its random initial value; training jointly with a projected-associator objective degrades the commutator residual roughly $100\times$ relative to isolated training, while a closure objective does not. The best-supported diagnosis is a genuine, structural conflict between the commutator-explanation objective and the associator/GJI family, mediated through the law’s parameters — not insufficient capacity, not gradient starvation, not the projector search being hard.

TABLE 1. Block B ablation matrix ($n = 16$, rank 4, cp-rank 4, 400 steps).

Regime	unexplained-rel	closure	associator
Isolated B only	0.000081	0.310	464.1
+ closure	0.000351	0.000004	74.6
+ associator	0.008486	0.111	0.000005
Joint (all)	0.005995	0.000479	0.000001
Frozen law, train projector	0.992488	0.0156	0.233
Frozen projector, train law	0.000000	0.738	580.9
Staged (competing, then B)	0.000465	0.217	141.1

3.3. **C — FINITE_BEALS_PROXY.** Renamed to make the scope limit explicit: this block reports finite nested-commutator Frobenius norms only, never a PsiDO-membership or microlocal-regularity claim. Adversarially-optimized projectors produce larger commutator norms (4.35) than random (3.84), smooth (2.85), or spatially-localized (2.00) hand-picked families, confirming the reported figures are not favorably cherry-picked.

3.4. **D — spectral snapping.** Tested on genuine near-projectors $P + \epsilon H$ (H Hermitian), not only exact ones. Rank recovery and a Davis–Kahan-style bound hold up to the point the spectral gap closes; an explicit counterexample ($\epsilon = 1.2$, gap closes from 1.0 to 0.51) reproduces rank misrecovery, confirming the gap condition is necessary.

3.5. **E — interscale subspace transport.** Three *independently* initialized and trained resolutions ($n = 12, 18, 24$), one frozen Gaussian-kernel lift operator per pair, principal angles against a random-subspace baseline and an interpolation baseline. All transported angles sit at 1.41–1.53 rad, essentially the maximum possible ($\pi/2 = 1.571$); the trained lift beats both baselines in only 2 of 3 pairs, by margins (0.02–0.15 rad) small relative to how close every condition already is to full orthogonality. No transport signal is found, consistent with block F’s independent basin-instability finding.

3.6. **F — rigidity.** The exact Hessian and the generalized Gauss–Newton approximation ($2J^\top J$) are computed and never conflated; GGN is confirmed PSD by construction, the exact Hessian is not (indefinite at a non-minimum, as expected). The infinitesimal gauge-rotation direction shows Hessian curvature -3.5×10^{-18} (numerically flat), confirming identifiability only modulo gauge — after a real methodological fix: the first version of the test loss was not actually gauge-invariant (a single-column construction, 0.87% discrepancy caught by the test itself), corrected by summing over all r columns (a Frobenius-norm quantity, exactly invariant under right-multiplication by a unitary). Basin stability across 3 seeds: all reach near-zero loss, but the pairwise maximum principal angle between converged subspaces is 1.55 rad ($\approx 89^\circ$) — essentially orthogonal, a genuine non-identifiability finding.

3.7. **G — n-ary closure.** Upgraded from 8–16 legacy trials to a 2000-sample empirical distribution plus adversarial gradient ascent (confirmed to find defects at least as bad as the best random sample). No interval/SOS certified bound was derived this pass; the claim is STATISTICALLY_VALIDATED_PASS, not PASS_CERTIFIED.

3.8. **H — associator constant.** The triangle-inequality bound $\|A(x, y, z)\| \leq 2\hat{M}^2\|x\|\|y\|\|z\|$ is never violated across 500 trials (300 random + 200 adversarial), but the maximum observed ratio is 0.452 — a 1.55 gap below the bound. The constant 2 is empirically far from sharp for this law family (mean $\cos(T_1, T_2) = 0.067$, near-uncorrelated rather than anti-aligned as the bound’s worst case assumes); no analytic tighter constant is derived, so sharpness remains OPEN.

3.9. **I — reduced tensor extraction.** Extraction-correctness only, as required. Two independent implementations (explicit loops; CP-exploiting einsum, including the cyclic average) agree to $< 10^{-10}$ at float64 and $< 10^{-4}$ at float32; an exact rational small case ($n = 2, r = 1$) matches a hand-expanded computation exactly, with zero floating point.

3.10. **J — tensor interscale.** The gauge-invariant comparison methodology (raw, Procrustes-aligned, permutation-aligned, amplitude-ratio, reported separately) is a genuine improvement over the legacy’s single Gram-eigenbasis heuristic. Applied to block E’s real three-resolution experiment: no persistence signal.

3.11. **K — HOSVD compactness.** A real reduced tensor needs mode ranks $[3, 4, 4]$ (out of full rank 4) to reach 99% energy, versus $[4, 4, 4]$ for a norm-matched random-tensor null — modestly, not dramatically, more compact, and only in one of three modes.

3.12. **L — gauge canonicalization.** Residual gauge is reported explicitly by spectral-cluster size. A deliberately constructed degenerate case (eigenvalues 2, 1, 1) is correctly detected, and an *arbitrary* (not infinitesimal) unitary rotation within the 2-dimensional degenerate eigenspace is confirmed to produce a pointwise-different but subspace-identical basis (principal angle $< 10^{-8}$ despite Frobenius difference > 0.1) — the concrete demonstration of “canonical only modulo residual gauge.”

3.13. **M — persistent factorization.** The comparison tool itself required a real fix during development: an early version compared HOSVD subspaces via free-unitary Procrustes and silently reported independent random tensors as gauge-equivalent, because the unitary group acts transitively on same-size orthonormal frames — caught by the block’s own required negative control, fixed with principal angles. Applied to block E’s three real, independently-trained resolutions: rank needed is inconsistent across scales ($\{3, 3, 4\}, \{3, 4, 4\}, \{3, 3, 3\}$); mean pairwise-mode principal angle 1.01 rad; one mode (of nine compared) shows a near-exact match (1.5×10^{-8} rad) unexplained by this pass and left as an open anomaly rather than smoothed into a summary verdict.

3.14. **N — cyclic law and GJI.** Raw (pre-averaging) cyclic defect 4.60 versus symmetrized (post cyclic-averaging) defect 8.2×10^{-33} — a 31-order-of-magnitude gap confirming the near-zero symmetrized figure is a construction identity, never learned evidence. One exact, versioned GJI formula (full antisymmetrization of the associator over the six permutations of three arguments) is implemented twice independently and agrees to 2.8×10^{-16} ; a deliberate sign-mutation is confirmed detected. The GJI-ratio supremum is not shown to be bounded (adversarial maximum 5.98 against a mean of 0.43) and is left OPEN.

4. EXTENDING THE SWEEP: PILOT AND BROAD SCREENING

The original pass above used single-configuration or few-seed evidence throughout — explicitly not a swept grid. A follow-up session executed two real, hardware-verified sweep stages against blocks A, G, H, N (the four with GPU support added for this purpose) on the confirmed RTX PRO 5000 Blackwell laptop GPU.

4.1. **Pilot (mission Phase 3).** 96 cells (arity $\in \{3, 4\}$, $n \in \{12, 24\}$, rank $\in \{3, 6\}$, cp-rank $\in \{4, 8\}$, 3 seeds, CPU+CUDA), 0 failures, 945.5 s wall time. GPU was $2.97\times$ slower than CPU at this scale (mean 14.70 s/cell CUDA vs 4.94 s/cell CPU), consistent with the single-sample CPU/GPU parity finding of the companion software paper (a 19.1 ms CPU vs 835.1 ms CUDA single-case comparison). Block H’s associator ratio, evaluated across 96 independent configurations rather than the single config of Section 3, ranged from 0.130 to 0.957 of the triangle bound (mean 0.334) — broader evidence for the same NOT_SHARP verdict, not a contradiction of the earlier 0.452 figure. Block N’s GJI-ratio adversarial maximum reproduced at 4.65–5.92 across the 96 cells (original single-config value: 5.98), confirming that finding is robust across configurations rather than a single-run artifact.

4.2. **Broad screening, Stage S1 (mission Phase 4).** 320 cells extending n to 96 ($4\times$ the pilot’s range), 5 seeds, 0 failures, 6316.5 s (≈ 105 minutes) wall time — this answers the specific question the pilot raised (does GPU ever overtake CPU at larger n ?). It does not, anywhere in the tested range:

n	CPU mean (s)	CUDA mean (s)	CUDA/CPU ratio
12	9.44	30.64	$3.25\times$
24	9.31	30.80	$3.31\times$
48	9.83	34.50	$3.51\times$
96	7.90	25.09	$3.18\times$

The ratio is stable across an $8\times$ range in ambient dimension — not a transient small- n effect. The concrete scheduling consequence for any further work on these four blocks: route to CPU, not GPU, at least up through $n = 96$. Blocks B, C, D, E, F, I, J, K, L, M were not extended to support the GPU device parameter and remain untested for this question. Stage S2 (adaptive intensification of anomalous/boundary cells) and Stage S3 (certification-mode reruns with held-out seeds, float64/complex128, deterministic algorithms) were not executed — both remain open work.

5. FAIL-CLOSED COMBINATION AND FINAL STATE

No global certificate is computed by averaging. Blocks B, E, J, and M resolve to FAIL for their scientific (non-structural) claims; blocks A, I, L, N carry genuine EXACT/STRUCTURAL_IDENTITY components alongside empirical residuals; blocks C, D, G, H, K remain at EMPIRICAL_SCREENING_PASS or weaker, single-seed evidence. The resulting state is FAIL_CLOSED_DYNAMIC_EXPLANATION_GATE_NOT_ESTABLISHED (named after the first failing required gate encountered, per `evaluate_global_certificate`); the honest global label most consistent with the full evidence set is PASS_A_TO_N_PARTIAL_CERTIFICATION at best, pending human review, never full certification, and several blocks (B, E, J, M) currently sit in an explicit failed state rather than an unresolved one.

6. LIMITATIONS

This pass used a closure-only (block E/F/G) or two-term (block B ablation) training objective for tractability, not the legacy’s full ~ 16 -term weighted loss; Section 3’s negative findings for blocks B/E/J/M are established for the objectives actually tested, not proved to persist under the historical loss, though block B’s own ablation matrix shows objective choice materially changes outcomes. Section 4 extends coverage to 416 total sweep cells across blocks A/G/H/N only — the other ten blocks remain single-configuration evidence. Track T is entirely out of scope, by the strict track-separation policy this program enforces.

A follow-up literature search (`docs/research/novelty_matrix_v5.md`) has since been performed for three of this program’s own methodological claims (the fail-closed typed-gate taxonomy itself, the legacy-lineage reclassification methodology, and the capacity-versus-deployment diagnosis behind block B’s ablation matrix). It found genuine, specific prior art in each case — a directly-named “fail-closed certification protocol” in an unrelated deployment domain, general ML provenance tooling for the lineage methodology, and a strong match to D’Amour et al.’s underspecification literature for block B’s capacity-versus-deployment phenomenon — so none of the methodological corrections described in this paper (Procrustes-on-subspaces, the gauge-invariant loss fix, or the typed-gate taxonomy itself) should be read as claimed novel. Every verdict in that search is recorded as PENDING_HUMAN_REVIEW: no automated process in this program self-approves a novelty verdict.

INDEPENDENT RESEARCHER, APIZACO, TLAXCALA, MEXICO